UALBANY

Electrocardiogram Signal–Based Cardiac Abnormality Detection

Dataset

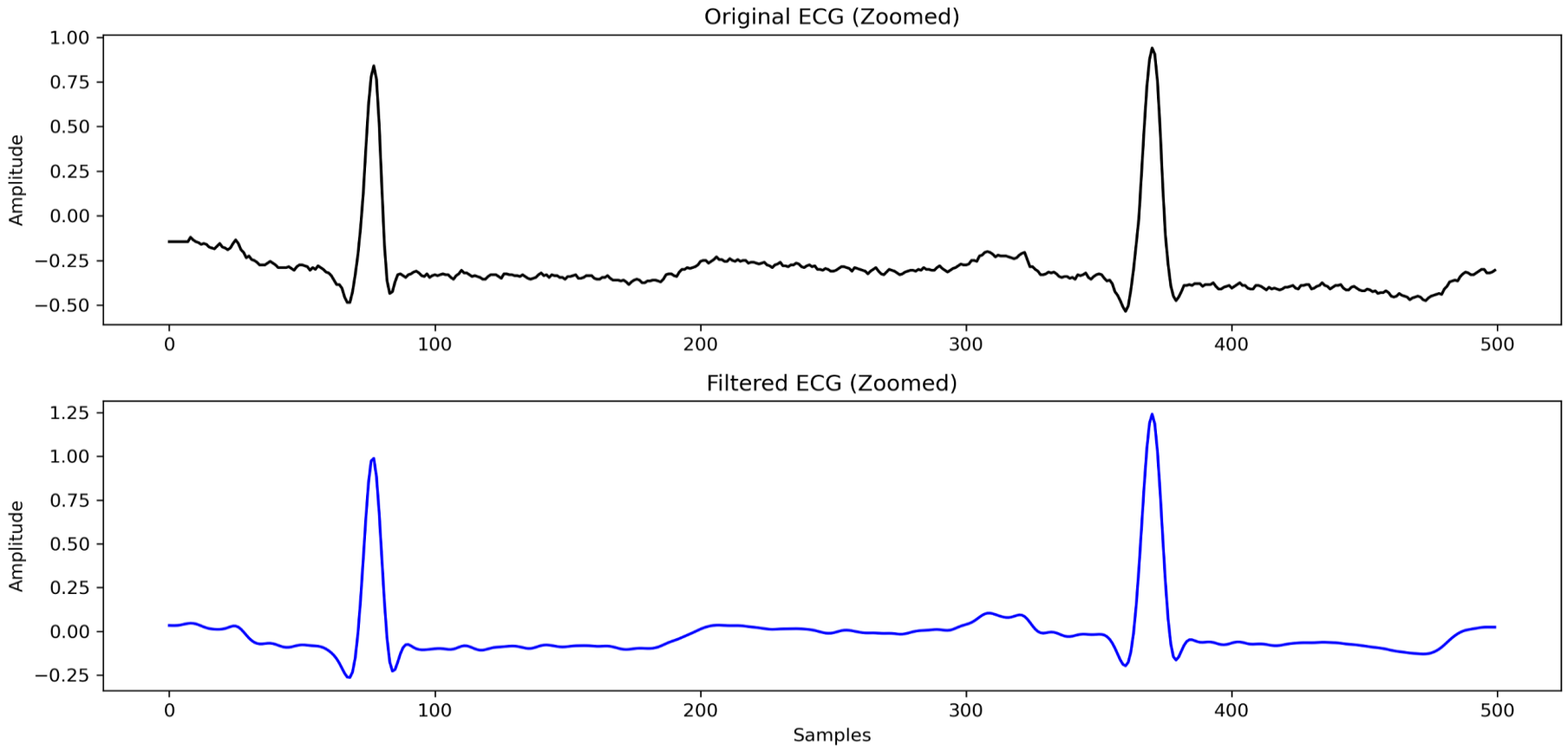
MIT-BIH Arrhythmia Database

The dataset contains 48 ECG recordings from different patients.

Each recording is approximately 30 minutes long and sampled at 360 Hz

Data Processing

Data Processing



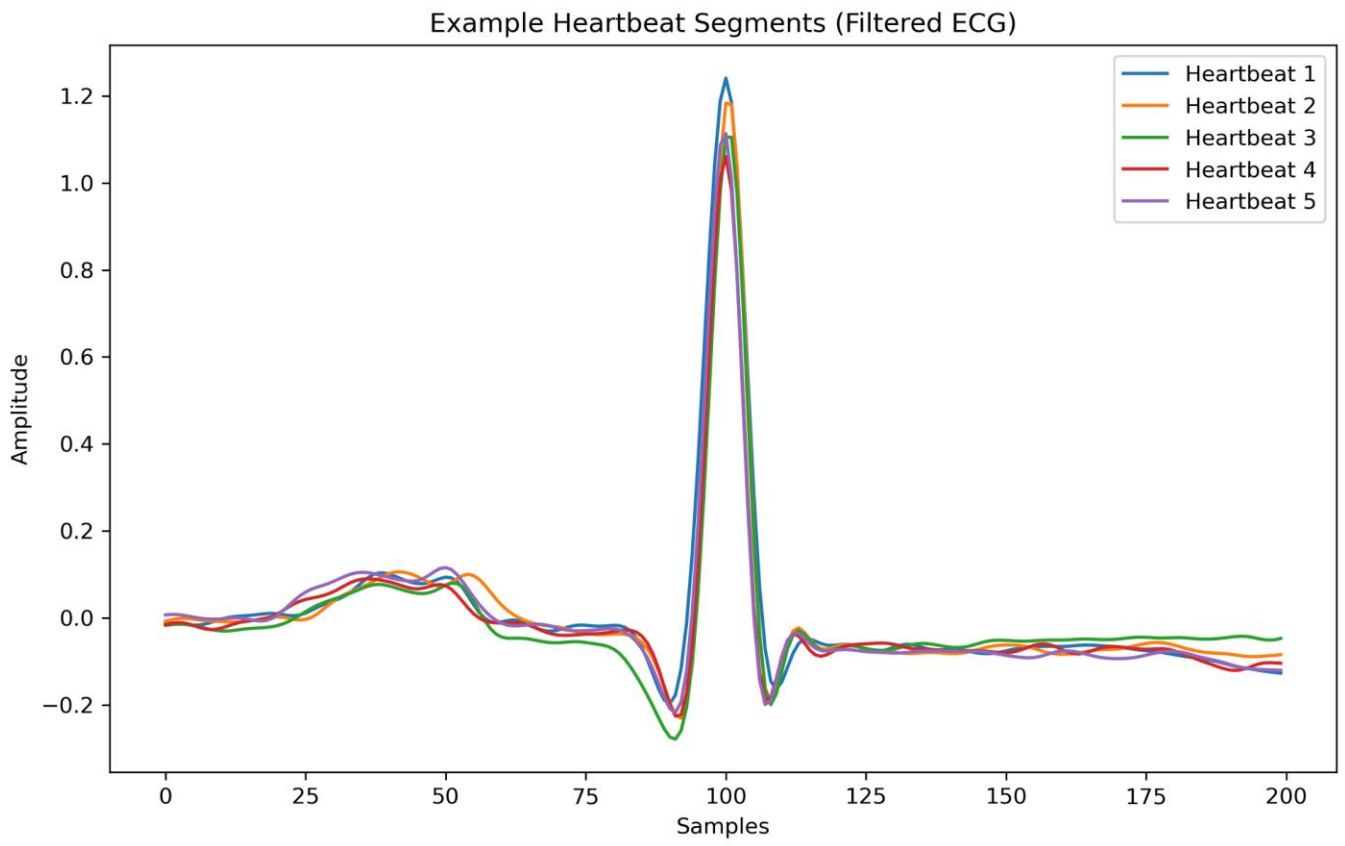
Noise in ECG signals
- **Baseline wander**
- **Powerline interference**
- **High-frequency noise**

Filtering Method
- **Bandpass filter (0.5–40 Hz)**
- **Remove noise**
- **Preserve ECG features**

Heartbeat Segmentation

The filtered ECG signal is segmented into individual heartbeats using the annotated R-peak positions.

Each heartbeat segment:
- centered around the R-peak
- contains 200 samples
- includes P wave, QRS complex, and T wave



Standardization

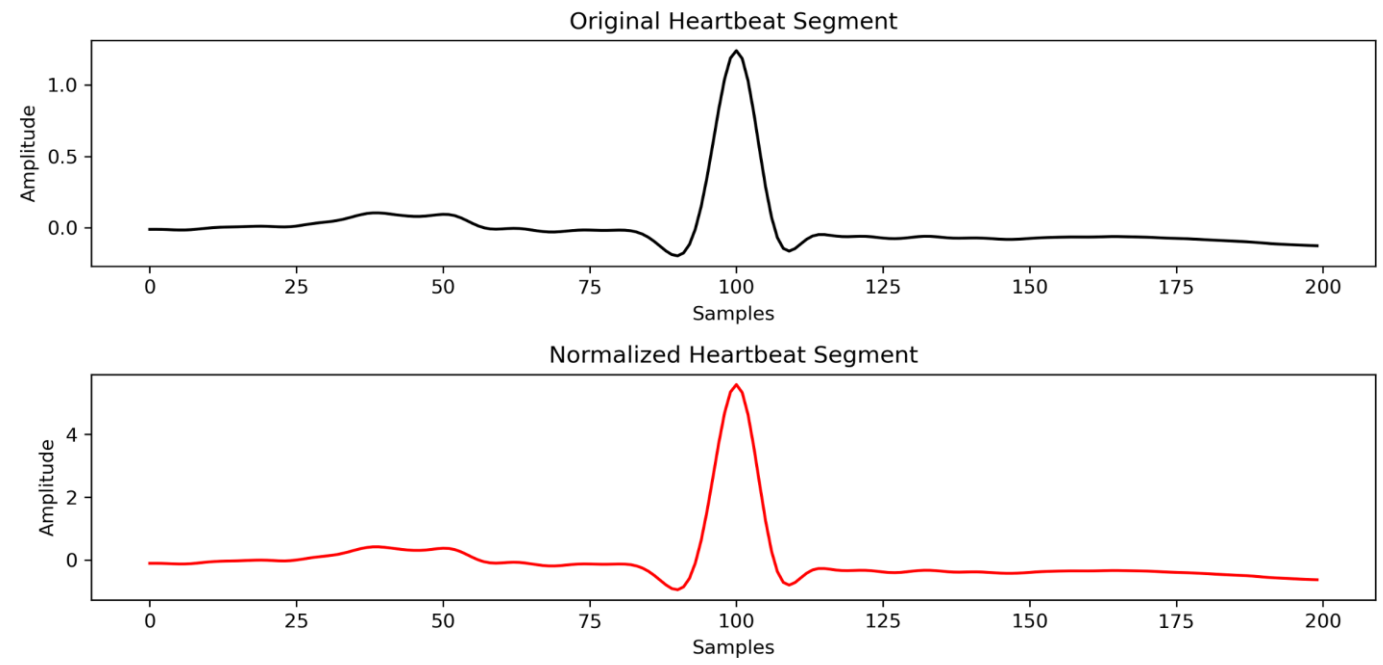
Standardization transforms the signal using:

$$x' = \frac{x - \mu}{\sigma}$$

μ : mean of the signal
σ : standard deviation

This operation:

- **centers the signal around zero**
- **rescales the amplitude**
- **preserves the waveform structure**



Why 1D CNN for ECG Arrhythmia Detection?

1D Temporal Structure

ECG is a 1-D time series. Conv1D kernels slide along the time axis, directly matching how P, QRS, and T-wave features are sequentially embedded in the signal.

Automatic Feature Extraction

No hand-crafted features needed. The network learns R-peak shape, beat morphology, and rhythm patterns end-to-end from raw windowed samples.

Efficient & Lightweight

Only ~445K parameters across 3 conv blocks. Each beat is a 200-sample window — fast enough for real-time clinical inference.

Handles Class Imbalance

Weighted cross-entropy (1:8) penalises missed arrhythmias more heavily, so the model learns rare patterns that a PCA+SVM pipeline often loses.

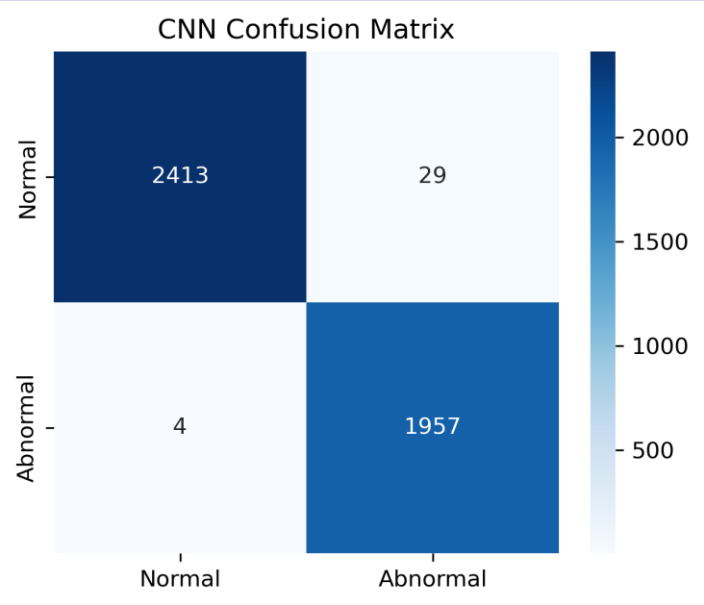
1D CNN Model Architecture

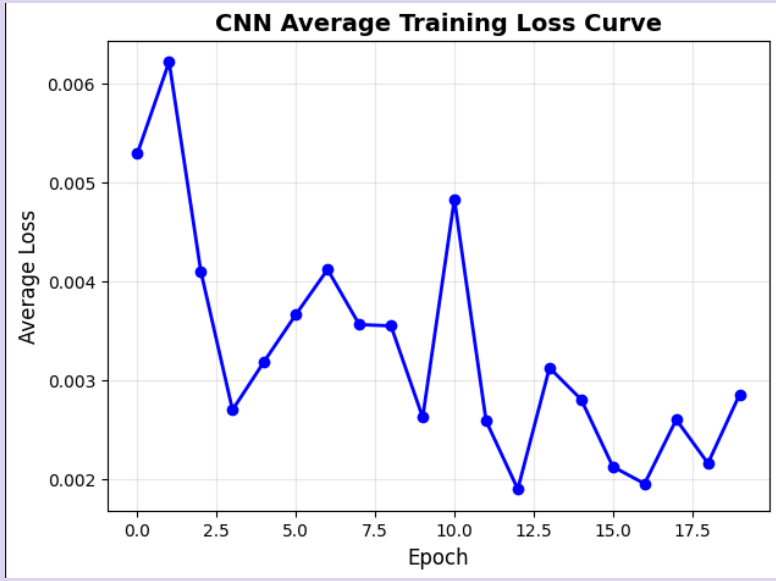
Layer	Input Shape	Output Shape	Param #
Conv1d	(None, 32, 200)	(None, 32, 200)	192
BatchNorm1d	(None, 32, 200)	(None, 32, 200)	64
ReLU	(None, 32, 200)	(None, 32, 200)	0
MaxPool1d	(None, 32, 200)	(None, 32, 100)	0
Conv1d	(None, 32, 100)	(None, 64, 100)	10,304
BatchNorm1d	(None, 64, 100)	(None, 64, 100)	128
ReLU	(None, 64, 100)	(None, 64, 100)	0
MaxPool1d	(None, 64, 100)	(None, 64, 50)	0
Conv1d	(None, 64, 50)	(None, 128, 50)	24,704
ReLU	(None, 128, 50)	(None, 128, 50)	0
MaxPool1d	(None, 128, 50)	(None, 128, 25)	0
Flatten	(None, 128, 25)	(None, 3200)	0
Linear	(None, 3200)	(None, 128)	409,728
ReLU	(None, 128)	(None, 128)	0
Dropout	(None, 128)	(None, 128)	0
Linear	(None, 128)	(None, 2)	258

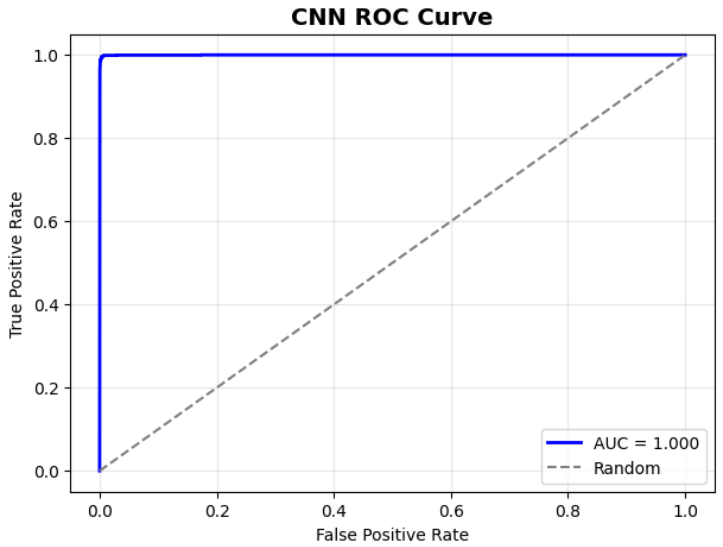
Raw ECG Beat (200 samples) ▶ **Conv1D Blocks (32 → 64 → 128)** ▶ **FC Layer (128 units)** ▶ **Output (Normal / Arrhythmia)**

CNN Model Results & Analysis

99.25%	98.54%	99.80%	1.000
Accuracy	**Precision**	**Recall**	**AUC-ROC**
(4370 / 4403 correct)	*Low false alarm rate*	*Only 4 missed arrhythmias*	*Perfect discrimination*





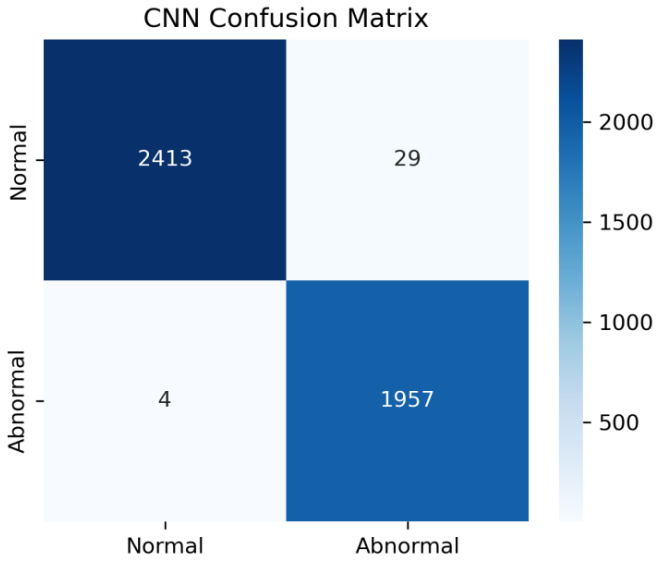


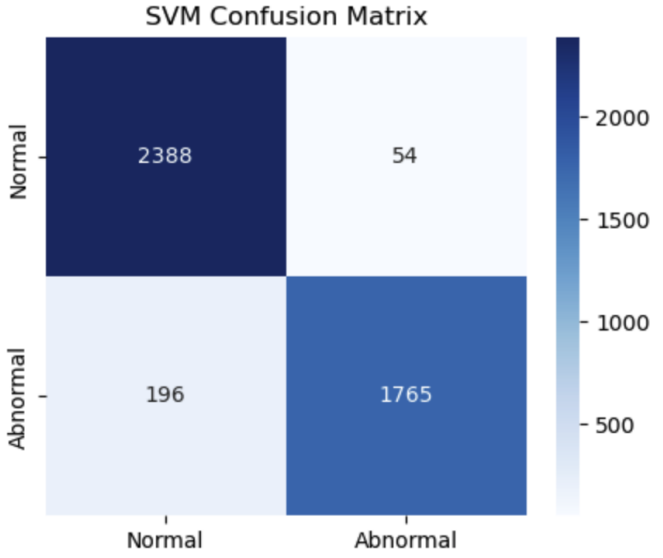
✓ **4 FN** — only 4 arrhythmias missed out of 1961
✓ **29 FP** — 29 false alarms from 2442 normal beats

⚠ **Oscillating loss** — caused by 1:8 class weight & small batch size
✓ **Downward trend** confirms model is converging over 20 epochs

✓ **AUC = 1.000** — perfect separation of classes
✓ **Curve hugs top-left** corner across all decision thresholds

Model Results

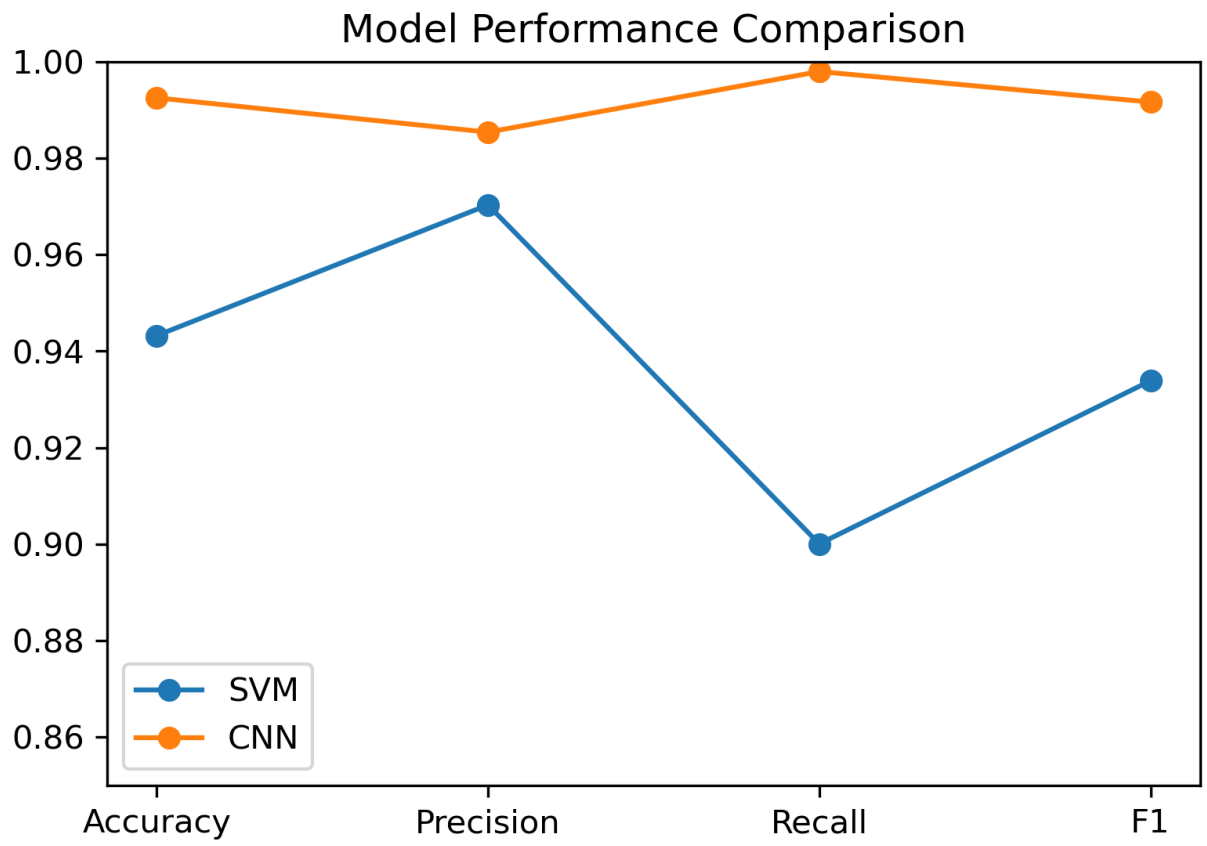




Confusion Matrix Comparison

- **CNN significantly reduces false negatives**
- **SVM misses many abnormal cases**
- **CNN achieves much higher recall**
- **CNN is more suitable for ECG abnormality detection**

Model Results

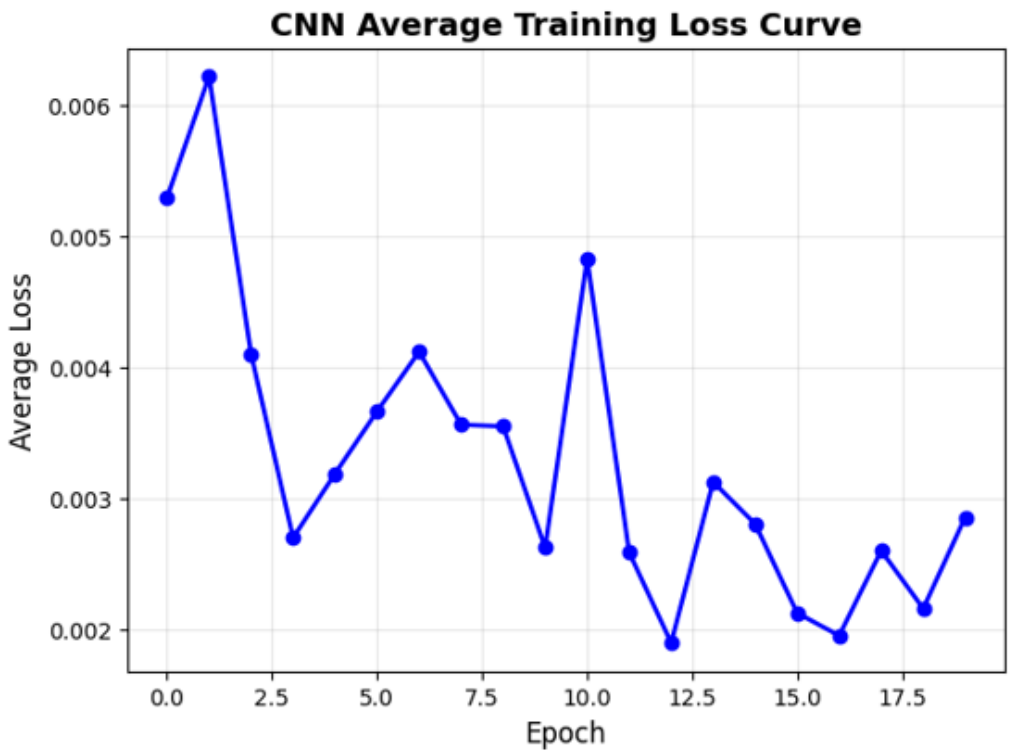


Model Performance Comparison

- CNN outperforms SVM across all evaluation metrics
- The most significant improvement is observed in recall
- SVM shows lower recall, indicating missed abnormal cases
- CNN is more suitable for ECG abnormality detection

Model Results

Training Loss Curve



- The training loss decreases rapidly in early epochs
- The model quickly learns useful features
- The loss gradually stabilizes after several epochs
- This indicates good convergence and stable training
- Note: Only training loss is shown, validation loss is needed to assess overfitting this is due to the 80/20 data split